

Lab 11 - Voiceover, Music, Captions and the Final Export Ladder

Course: Create Short Video Film using AI (TGS-2020505925) | Version v12.0 | 27 September 2026 Topic 04: AI Video Generation, Animation and Video Analytics Outcome: ELO4 | In-class time: 30 minutes of scheduled practical time

Competency	Statement
Ability A5	Design and apply machine-learning based methods for object detection, object tracking and activity recognition
Ability A6	Design and apply video analytics algorithms for high-level video analytics tasks
Knowledge K9	Object segmentation, detection and recognition
Knowledge K10	Activity tracking, generative models, scene understanding and event discovery

Scenario

The last ten seconds of The Last Ferry are picture-locked. You finish them: validate the subtitles against the house caption rules, build and measure a placeholder mix (voiceover cues, a music bed at the brief's tempo, rain and a ferry horn, with the music ducked under the voice), and export a 16:9 master, a 9:16 Reel and a 4:5 feed version that follow MEI using her raincoat's segmentation mask.

What you will produce

A caption file that passes every rule, an audio report (peak, RMS, voice-over-music margin), and three exported masters with burned-in captions - muxed to H.264/AAC when FFmpeg is installed.

Tools: Python 3 | opencv-python | NumPy | optional FFmpeg | optional: a voice, music and editing tool

Environment

Install these once, before class if you can - the lab itself needs no network access after the dependencies are present:

```
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\Activate.ps1
pip install --upgrade pip
pip install "opencv-python>=4.10" numpy
python3 -c "import cv2, numpy; print(cv2.__version__, numpy.__version__)"
```

Required: opencv-python>=4.10, numpy, ffmpeg (optional, for muxing audio).

API currency. Everything taught here runs on the base opencv-python wheel plus NumPy - no contrib modules and no downloaded model weights. SSIM is implemented in NumPy because cv2.quality is contrib-only. Generative steps are optional: every lab supplies SIMULATED reference media so it can be completed offline, and asks you to record the model ID and date if you generate your own.

Workflow

1. Validate captions against the rules
2. Fix the caption file
3. Build and measure the mix
4. Reframe and export each format
5. Verify the delivered files

Files in this folder

Path	What it is
reference/picture-lock.mp4	10-second picture lock of the last three shots (SIMULATED clip)
data/captions.srt	Subtitle file with five deliberate faults
data/caption-rules.json	Line count, line length, reading speed, duration and gap rules
data/vo-cues.json	Voiceover cue sheet with audio tags
data/music-brief.json	Music prompt, tempo, ducking and level targets
data/export-profiles.json	16:9, 9:16 and 4:5 profiles with safe areas
finish.py	Runnable script - see the steps below
PROMPTS.md / PROMPTS.pdf	The reusable prompt and specification pack
out/	Everything you produce (created on first run)

Provenance. Every image and clip in reference/ is either a SIMULATED classroom illustration of *The Last Ferry*, rendered deterministically with OpenCV and labelled on the frame, or an AI-generated reference copied unchanged with a PROVENANCE.md beside it. Nothing in this folder may be relabelled: a simulated stand-in is never presented as model output or as a photograph.

Steps

1. Read data/vo-cues.json. Each line carries an audio tag such as [out of breath] or [stunned whisper] - that is how delivery is directed line by line.
2. Read data/music-brief.json: the music prompt, 86 BPM, the ducking depth, the peak ceiling and the minimum voice-over-music margin.
3. Validate the captions. The check exits with status 1 and lists five faults.

```
python3 finish.py captions --srt data/captions.srt --rules data/caption-rules.json
```

4. Copy the file to out/captions-v2.srt and fix it: delete the stray 'Four minutes.' cue, shorten cue 3 to 'No sender. / Just a tag, folded shut.' on two lines, and renumber. Re-run the check on your file until it passes.

```
python3 finish.py captions --srt out/captions-v2.srt --rules data/caption-rules.json
```

5. Build and measure the placeholder mix.

```
python3 finish.py audio --srt out/captions-v2.srt --out out
```

6. Read the audio report: peak below -1 dBFS, and voice at least 6 dB above the music while the voice is active. Change duck_db to 0 and rerun: record the new margin, then restore 9.
7. Export everything.

```
python3 finish.py all --srt out/captions-v2.srt --out out
```

8. Read the export ladder: the 9:16 Reel keeps about 31.6% of the master and follows MEI's raincoat mask; the 4:5 keeps about 45%. With FFmpeg each file is muxed to H.264 video and AAC audio.
9. Open out/reels_9x16.mp4 (or _final.mp4). Check that the captions sit above the platform UI zone and that the SIMULATED label survived the crop.
10. Write out/delivery-checklist.md: captions pass, peak and margin, sizes and frame counts, subject in frame, provenance label present, AI-content disclosure set on upload.
11. Optional extension: replace the placeholder with a real voiceover and music track generated from the cue sheet and brief, then re-measure.

Verify

Your caption file passes every rule; the mix peaks at or below -1 dBFS with a voice margin of at least 6 dB; three exports are written with 240 frames each.

Expected outputs

- out/captions-v2.srt passing the check
- out/mix-placeholder.wav and out/audio-report.json
- out/master_16x9.mp4, reels_9x16.mp4, feed_4x5.mp4 (+ _final.mp4 with FFmpeg)
- out/export-report.json and out/delivery-checklist.md

Evidence checklist

Submit all of the following. Each item is something an assessor can look at.

- ☐ Failing caption check and your corrected caption file passing
- ☐ Audio report, and the margin with ducking switched off
- ☐ Export report with sizes, frame counts and kept percentages
- ☐ Delivery checklist including provenance and AI-content disclosure

Troubleshooting

Symptom	What to do
'Fix the caption problems before exporting'	Burned-in captions cannot be edited; fix the SRT first.
'ffmpeg not found: video only'	Install FFmpeg (brew install ffmpeg / winget install ffmpeg) or mux in your editor.
Captions overlap the platform buttons	Raise safe.bottom for that profile in export-profiles.json.
The vertical crop loses MEI	follow_subject must be true; check the coat colour range finds her.

References used in this lab

- [S6] Youri van Hofwegen, 'EXACTLY How to Make an AI Short Film (Full Workflow)', YouTube, 1 Jun 2026, youtube.com/watch?v=0v534yAyhwg - *Full video, 16:49*.

Reference-first principle (stop giving the model freedom); three reference types - characters, locations, props; casting attributes (genre, budget, era, archetype, identity, physique, details, outfit); a second, damaged-state location made by editing the first ('change nothing else'); director controls (genre, camera movement, speed ramp, duration, aspect ratio, audio); per-character emotion and intensity; the multi-shot prompt framework with dialogue in quotation marks; video reference versus start/end frame (mood carries over); three generations for dense action shots; assembly in a timeline editor.

- [S7] Mira AI, '6 AI Video Tools I'd Use To Make An AI Short Film', YouTube, 14 Jun 2026, youtube.com/watch?v=Whh8vCqboxQ - *Full video, 10:04*.

A multi-tool workflow: an LLM to plan the story shot by shot, a cinematic image model for character sheets (front, side, back, close-ups) and multi-angle environment grids, a design canvas for the storyboard with keyframe placeholders, a video model for motion, an AI music generator (instrumental, mood and tempo), and an editor; the six-point prompt framework (subject, action, environment, camera, style, constraints); timeline prompting; video extension to pass the previous clip; generations capped at about 15 seconds.

- [S8] Thomas Creates, 'How to Start Making AI Short Films in 2026', YouTube, 18 Sep 2026, youtube.com/watch?v=69_EJD4FEPg - *Video description and chapters, 12:42*.

Consistent characters, props and locations; animating cinematic shots; carrying the same lighting and look between scenes; generated narration; assembling the film.

- [S10] Youri van Hofwegen, 'How to Make Your First AI Movie (Full Guide)', YouTube, 11 Jun 2026, youtube.com/watch?v=fs5S867VQzg - *Video description, 17:03*.

Generating the story; building consistent characters and locations; JSON prompting; asset management; video references for connected scenes; editing; fixing voice continuity across clips.

- [S11] ElevenLabs, 'How to Make Your First AI Short Film (Full Tutorial)', YouTube, 9 Aug 2026, youtube.com/watch?v=47DH_VUa67A - *Chapters, 31:13*.

Explore cheaply with faster image models before committing credits; prompt structure across style, subject, objects and negative prompting; a character reference sheet; generate the voiceover FIRST and build the storyboard to match it; direct delivery line by line with audio tags; still frames before video; start and end frames; generate at lower resolution and upscale only the takes you keep; background music; sound effects; edit and trim on a timeline; regenerate with a new character.

© 2026 Tertiary Infotech Academy Pte Ltd. UEN: 201200696W. Course material for TGS-2020505925, version v12.0.